

Response Letter #1

Dear Editors and Reviewers,

Thank you for your letter and the constructive feedback on our manuscript. We sincerely appreciate the time and effort you have dedicated to evaluating our work and providing insightful suggestions to improve its quality.

We are grateful to the reviewers for their thorough and thoughtful comments, which have helped us refine the methodology, clarify key arguments, and strengthen the overall presentation of the paper. We have carefully addressed all concerns raised in the review reports and incorporated the suggested revisions into the revised manuscript.

The major revisions of this manuscript are listed as following:

Structural Adjustments

- The manuscript now follows the sequence: **Introduction, Results, Discussion, and Methods**, aligning with the journal's formatting guidelines.
- A **Nomenclature** section has been added at the beginning of the text to clarify terminology for readers.

Results

- Expanded analysis on the differences in actual contact area versus pressure between two specimen groups,
- Include comparisons with existing prediction models.
- Results related to single-surface interfaces have been **moved** to the **Supplementary Discussion** to streamline the main text.

Discussion

- Simplified conclusions and focused on highlighting **limitations** and **further researches**.

Methods

- Added details on surface generation procedures.
- Clarified the theoretical framework and computational methods for generating the training dataset.

Figures

- **All** figures have been **re-created** to meet publication standards (resolution, font size, and line weight adjusted). **All** figure captions were **rewritten** for clarity.

Supplementary Methods now include:

- Experimental principles and uncertainty calculations for TCR measurements.
- Step-by-step protocols for surface generation and fractal parameter determination.
- Descriptions of the CMY empirical model and ML prediction model for reproducibility.

Supplementary Discussion provides:

- A systematic comparison between our approach and FEM-based methods.
- In-depth explanations for the observed differences in contact area-pressure trends between specimen groups.
- Detail analysis for the difference in real contact area changing with contact pressure.

Text Revisions

- The entire manuscript has been proofread for consistency, with improved flow between sections and reduced redundancy.

A point-by-point response to each comment is provided below, along with the corresponding changes highlighted in the updated manuscript.

Reviewer #1:

The manuscript COMMS-25-0150-T entitled “What surface characteristics truly affect thermal contact resistance—An interpretability study based on deep learning and convolutional neural networks” presented a deep learning model based on convolutional neural networks for the prediction of thermal contact resistance. The effects of surface characteristics on thermal contact resistance have also been studied. The models have also been validated with the experimental data. The work presented in the manuscript is very relevant to the thermal management of high-performance devices for effective heat dissipation. The manuscript is well written and well presented. However, after careful reading of the manuscript, there are major comments and suggestions. Therefore, I recommend accepting this manuscript if the authors convincingly address the considerations below:

1. Please provide nomenclature for the symbols and abbreviations used.

We sincerely thank the reviewer for pointing out the need to clarify the symbols and abbreviations used in the manuscript. To enhance clarity and accessibility, we will include a dedicated Nomenclature section in the revised version. This section will systematically list all symbols, abbreviations, and acronyms used throughout the paper, along with their corresponding definitions (where applicable).

2. All figure captions are very detailed. Those should be written in brief while the details about the figures may be discussed in the body text citing the figure.

We thank the reviewer for highlighting the need to streamline our figure captions. In response, we will revise all figure captions to ensure they are succinct and primarily serve to identify the figure’s content and purpose. The detailed explanations and interpretations of the figures, which are currently included in the captions, will be moved to the main text.

3. The theoretical model used for data generation is an important part of the present study; therefore, it should be explained in detail in the manuscript. The supplement also does not have sufficient details of the model.

We appreciate the reviewer's comment regarding the need for a detailed explanation of the theoretical model used for data generation in our study. In response to this feedback, we have incorporated the theoretical model into the **Methods** section of the manuscript.

Given that the theoretical model is not original to this work, we have chosen to present only the essential computational formulas necessary for result reproduction, rather than detailing the full derivation process. We believe these provided formulas will enable readers to perform the calculations independently and understand the underlying principles without the need for extensive derivation.

Additionally, specific details concerning the generation of fractal surfaces, which are part of our modeling process, have been included in the **Supplementary Methods 3**. This offers further insight into the procedural aspects that complement the main text.

Our aim is to balance providing sufficient detail for transparency and reproducibility with maintaining clarity and focus within the main manuscript. We hope these revisions address the reviewer's concerns and facilitate a better understanding of our methodology.

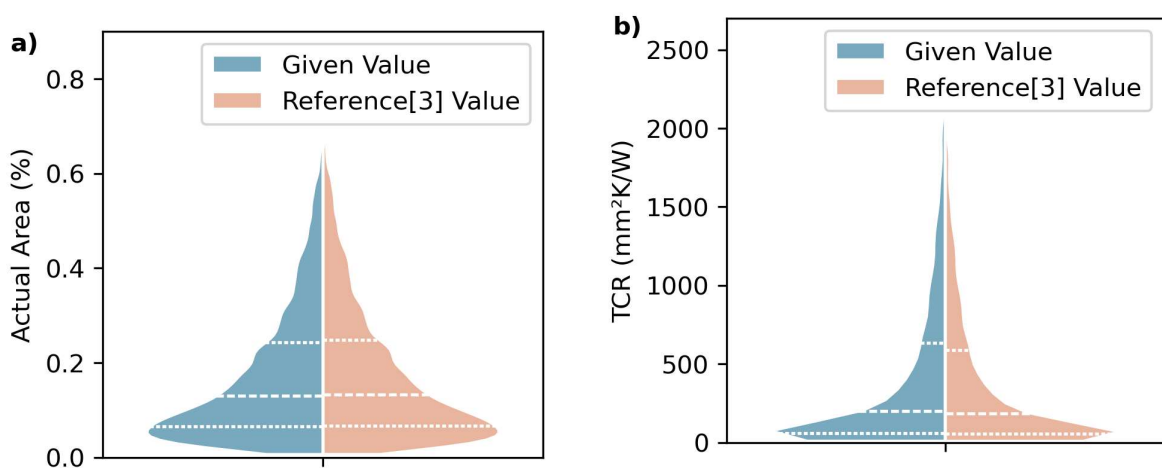
4. Please provide the reference of the thermomechanical properties of 316 stainless steel.

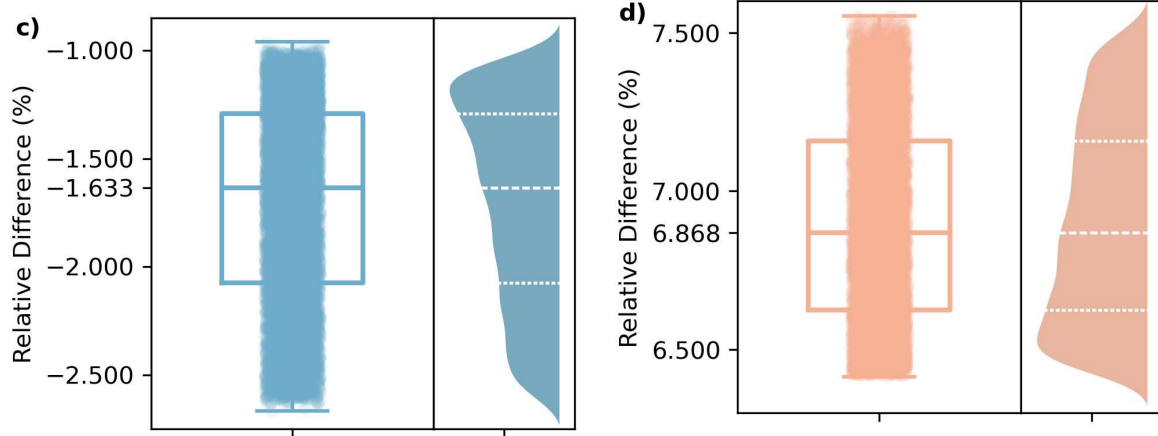
The microhardness data, converted from Vickers hardness, were obtained from Reference [1]. Data for elastic modulus, Poisson's ratio, and thermal conductivity were provided by the specimen supplier. The thermal conductivity values are consistent with those reported in Reference [2], while the elastic modulus and Poisson's ratio differ slightly from the values in Reference [3] ($E = 198 \text{ GPa}$, $\nu = 0.29$).

Response Figure 1 presents a comparison of the actual contact area and contact thermal resistance derived from theoretical calculations using these two sets of parameters in our model. As shown in **Response Figure 1 a)** and **b)**, the overall distributions of the results are not significantly affected by the different parameter values. Using literature values leads to slightly larger actual contact areas and smaller contact thermal resistance values compared to our calculations. Specifically, the actual contact area exhibits a relative difference of -2.5% to -1% (**Response Figure 1 c)**), while contact thermal resistance shows a discrepancy of approximately 6.868% (**Response Figure 1 d)**). The relative difference is calculated using the following equation:

$$\text{Diff} = \left(1 - \frac{\text{Val}_{\text{ref}}}{\text{Val}_{\text{given}}} \right) \times 100\%$$

We contend that this systematic discrepancy—either a uniform increase or decrease—does not undermine the core conclusions of this paper.





Response Figure 1. Comparison of results obtained using the elastic modulus and Poisson's ratio provided in this paper versus values from Reference[3] via the theoretical model: **a)** Comparison of violin plots for actual contact area. **b)** Comparison of violin plots for contact thermal resistance. **c)** Distribution of relative differences in actual contact area. **d)** Distribution of relative differences in contact thermal resistance.

- [1]. Svahn, F., Mishra, P., Edin, E., Åkerfeldt, P. & Antti, M. L. Microstructure and mechanical properties of a modified 316 austenitic stainless steel alloy manufactured by laser powder bed fusion. *Journal of Materials Research and Technology* 28, 1452-1462 (2024). <https://doi.org/10.1016/j.jmrt.2023.12.063>
- [2]. Y. S. Touloukian, R. W. P., C. Y. Ho, P. G. Klemens Thermal Conductivity Metallic Elements and Alloys. Vol. 1 (Springer, 1970).
- [3]. Hammond, J. P., Ratcliff, L. T., Brinkman, C. R., Moyer, M. W. & Nestor, J. C. W. Dynamic and static measurements of elastic constants with data on 2 1/4 Cr--1 Mo steel, types 304 and 316 stainless steels, and alloy 800H. (United States, 1979).

5. How is TCR calculated experimentally, and what are the experimental uncertainties? Please discuss.

We thank the reviewer for their inquiry regarding the experimental calculation of TCR and the associated uncertainties. The detailed methodology and uncertainty analysis are provided in **Supplementary Method 2-3**.

To summarize, the TCR is experimentally determined by the steady-state method based on ASTM D5470. The uncertainty is 5.79%-8.81% for our experiment.

We appreciate the reviewer's attention to these important aspects of our experimental methodology.

Response Table 1. Uncertainties for current experiment.

U_T	U_x	$\frac{U_k}{k}$	T_{m1}	T_{m2}	T_{s1}	T_{s2}	x_m	x_s	$\frac{U_R}{R}$
0.01 K	0.02 mm	3%	50-65°C	35-30°C	40-60°C	30-40°C	2mm 5mm 8mm	5 mm 30 mm 55 mm	5.79%- 8.81%

6. Only contact pressure and roughness are considered as parameters affecting thermal contact resistance, not mechanical properties. Therefore, results of the present study will not be valid for other materials? Please explain.

We thank the reviewer for their insightful comment regarding the parameters considered in this study. Under the current framework, predicting TCR for new materials or operating conditions would require generating new datasets and retraining the model. This limitation is indeed inherent to the present work, as discussed in detail in the **Discussion** section of the manuscript. However, we note that the costs associated with this retraining process are manageable within practical engineering contexts, as detailed in the **Supplementary Discussion 1**. The modular design of our approach allows for systematic extension to include additional parameters, such as mechanical properties, in future studies.

7. Line 320: “Both groups exhibit a surface roughness of approximately 0.8.” What is this roughness: RMS, Ra, or Rz? And what is the unit, in microns or in nanometers?

We sincerely thank the reviewer for their meticulous review and for prompting this clarification. The surface roughness value of 0.8 refers to the arithmetic mean roughness (Ra), and the unit is microns (μm).

Revision content (together with other revision):

Both groups exhibit an Ra of approximately 0.8 μm .

We also identified the same issue elsewhere:

Revision content (together with other revision):

...experimental tests are conducted on two sets of 316 stainless steel specimens with an ~~surface roughness~~ Ra of approximately 0.8 μm .

We appreciate the reviewer’s attention to detail, which strengthens the precision of our work.

8. Contact thermal resistance is measured experimentally under pressures ranging from 1 to 4.55 MPa, while prediction is for 1 to 10 MPa. Why not in the same pressure range?

We appreciate the reviewer’s thoughtful question regarding the discrepancy between the pressure ranges used in the experimental validation and those covered by the model prediction.

Our deep learning model was developed and trained based on a dataset generated from the surface fractal model, which was originally validated in a previous study over the pressure range of **1–10 MPa**[1]. In that study, the fractal-based theoretical framework demonstrated good agreement with numerical simulations across this wider pressure range. Therefore, to maintain consistency with the data generation process and to explore the broader predictive capability of our model, we selected the **1–10 MPa** range for training and evaluation.

On the other hand, the experimental validation was conducted using our current testing setup, which has a practical pressure limit of approximately **5 MPa** due to hardware constraints. As

a result, the experimentally measured thermal contact resistance (TCR) values are limited to the **1–4.55 MPa** range.

This inconsistency arises not from any flaw in the methodology, but rather from the temporal sequence of model development and experimental validation — the model was built and trained before the experimental setup was finalized. We believe that despite this mismatch, the experimental results within the overlapping pressure range sufficiently validate the model's predictive accuracy under realistic conditions.

[1]. Ma, C., Zhao, L., Shi, H., Mei, X. S. & Yang, J. A geometrical-mechanical-thermal predictive model for thermal contact conductance in vacuum environment. Proceedings of the Institution of Mechanical Engineers Part B-Journal of Engineering Manufacture 230, 1451-1464 (2016). <https://doi.org/10.1177/0954405415611358>

9. The percentage increase in contact area with pressure from 1-10 MPa is 0.02% to 0.25% for wire cutting specimens, while it is 0.02% to 0.4% for turning specimens. Explain why.

We sincerely thank the reviewer for raising this important point regarding the different percentage increases in contact area with pressure between turned and ground specimens.

To begin, we would like to clarify that **the predicted contact area values are theoretical estimates**, derived from our surface fractal model and multi-point contact mechanics framework. These values have not been experimentally validated in this study, and as noted in the manuscript (see Figure 1), the accuracy of these predictions is limited. Therefore, **the observed differences should be interpreted with caution** and may not fully reflect real-world behavior.

That said, based on the modeling results and our understanding of surface generation mechanisms, we offer the following explanation for the differing trends in contact area increase with pressure:

1. Influence of Surface Morphology on Contact Behavior

The difference in the rate of contact area increase is attributed to variations in **surface topography characteristics**, particularly the **shape and distribution of asperities**, which are influenced by the machining process:

- Turning typically produces surfaces with more **directional, defined, and sharper asperities** — often exhibiting a saw-tooth or cusp-like profile due to the geometry of the cutting tool and the feed motion. These sharper peaks result in higher stress concentrations at the contact points under applied pressure.
- Grinding tends to generate surfaces with more **randomly distributed, rounded, or crater-like asperities**. These asperities possess less acute peak geometries and are generally more uniformly distributed across the surface.

Even when both types of surfaces exhibit **similar arithmetic average roughness (Ra)**, the **distribution of asperity heights and lateral spacing** can differ significantly. This leads to distinct mechanical responses under increasing pressure.

2. Hypothetical Illustration

To illustrate this concept, consider two idealized extreme cases:

- A turned surface has N asperities, each with height $2h$.
- A ground surface has $2N$ asperities, each with height h .

At low pressures, both surfaces may undergo primarily **plastic deformation**, resulting in similar small contact areas proportional to material hardness.

However, as pressure increases, **elastic deformation becomes dominant**, and the response differs:

$$f_e \propto a^{2-D}$$
$$\frac{a_1}{a_2} = \left(\frac{2f}{f}\right)^{1/(2-D)} = 2^{1/(2-D)} > 2, 1 < D < 2.$$

The taller asperities on the turned surface deform elastically over a larger area.

This results in a **greater overall increase in relative contact area** for turned surfaces compared to ground one under higher pressure.

In summary, even when R_a is controlled to be similar, the actual contact behavior can vary significantly due to differences in asperity shape and spatial arrangement introduced by different machining processes. Turned surfaces tend to have sharper peaks that deform differently under pressure, while ground surfaces, with their more rounded asperities, resulting in a greater increase in real contact area as pressure rises for turned surfaces.

We have added this discussion to both the **Results** section in main manuscript and the **Supplementary Discussion 3** to enhance the clarity and depth of our analysis.

Thank you again for your insightful comment, which has helped us better explain the underlying physical mechanisms in our study.

10. Line 336: A typo error; figure is written two times.

Thank you for your meticulous identification of the spelling errors in the manuscript. Such details are vital to maintaining the scholarly integrity of the paper. These contents have been moved to the **Supplementary Discussion 2**. Corresponding revisions have been made in the relevant sections accordingly.

11. Figs. 5 (d, e, f, g, k, l, m, n) have very small texts, so they are not visible.

We sincerely thank the reviewer for bringing the legibility issue in this Figure to our attention. We fully acknowledge that the text in panels (d, e, f, g, k, l, m, n) is currently too small and may hinder clarity. To address this, we have revised all affected panels to ensure all text (including labels, legends, and annotations) is now sufficiently large and readable. Considering these results serve as supplementary validations to our core predictive findings, we have strategically relocated the revised figures to **Supplementary Figure 1-2**.

12. To understand the significance of your study, it's essential to compare your results with the existing ML-based studies to estimate TCR/TCC in the literature.

We sincerely appreciate the reviewer's suggestion to benchmark our findings against existing ML-based studies on TCR or TCC estimation. This comparative approach indeed serves as an effective means to emphasize the originality and impact of our work.

To address this comment, we first acknowledge that, to the best of our knowledge, there are no existing ML models in the literature that directly tackle TCR/TCC estimation using the same material systems as those investigated in this study[1-3]. To address this gap, our ongoing research, which aims to establish such a framework, has been shared as a preprint on Chinaxiv[4] and is currently undergoing further refinement and peer review.

In response, we have incorporated a comparative analysis between the current model and the methodology described in our preprint. Results are exhibited in **Figure 2e-f**) in the revised manuscript and **Response Figure 2** together with other prediction models.

While this preliminary work provides contextual baseline insights, it is important to highlight that it remains under peer review and should be considered as a work in progress.

We firmly believe that the combination of this preprint and the present study collectively advances the application of deep learning techniques in TCR/TCC estimation, contributing both to the methodological foundation and the exploration of material-specific solutions.

- [1]. Vu, A. T., Gulati, S., Vogel, P.-A., Grunwald, T. & Bergs, T. Machine learning-based predictive modeling of contact heat transfer. *International Journal of Heat and Mass Transfer* 174 (2021). <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121300>
- [2]. Feng, Z. Y., Yan, J. T. & Gao, Y. W. Prediction of contact resistance between copper blocks under cyclic load based on deep learning algorithm. *Aip Adv* 12, 075009 (2022). <https://doi.org/10.1063/5.0095871>
- [3]. Ren, X.-J., Gou, J.-J., Dai, Y.-J. & Tao, W.-Q. Exploring thermal contact conductance between two contact solids by artificial neural network. *International Communications in Heat and Mass Transfer* 136, 106182 (2022). <https://doi.org/10.1016/j.icheatmasstransfer.2022.106182>
- [4]. Man Zhou, Pei Yao Guo, Zhuo Yan He, Ping Zhang & Tao, L. The Thermal Contact Resistance Dataset and the Artificial Intelligence-Driven Prediction of Thermal Contact Resistance in Multi-material Systems. *Chinaxiv e-prints*, ChinaXiv:202504.200193 (2025). <https://chinaxiv.org/abs/202504.00185>

13. Discuss the limitations of the present study and the techniques used.

We appreciate the reviewer's insightful comment regarding the limitations of our study. Indeed, as this research represents an early effort in its field, it inevitably encounters several constraints that merit discussion. We have addressed this thoroughly in the **Discussion** section of the manuscript. Building on your feedback, we have further articulated the limitations of our research and the techniques employed as follows:

- **Acquisition of datasets:** The surface patterns generated via fractal theory exhibit excessive homogeneity in the training dataset, which may lead the neural network to overfit to these specific surface configurations. This could result in suboptimal feature extraction processes, potentially causing the model to either overemphasize certain surface characteristics or overlook others during feature selection.

- **Selection of description sets:** The current description set is narrowly confined to pressure parameters and surface topography metrics, restricting the model's predictive capability to a single material type. Generalizing this framework to other materials or operating conditions would necessitate comprehensive dataset regeneration and model retraining.
- **Construction of deep learning architectures:** A related limitation arises from the model architecture's separate processing of pressure inputs and surface characteristics, which may impair its ability to capture synergistic effects between pressure conditions and surface feature interactions.
- **Experimental validation:** this study's empirical verification was limited to contact thermal resistance measurements, with no direct validation of predicted actual contact area values. Consequently, the model's contact area predictions should be interpreted as approximate estimates rather than definitive measurements.
- **Visualization of models:** the research lacks methodological explanations for the class activation mapping (CAM) visualizations generated by the deep neural network, hindering interpretability of the model's decision-making rationale.

We acknowledge these limitations and believe they provide valuable directions for future research.

14. Discuss the areas for further studies, highlighting the issues that remain to be explored.

We thank the reviewer for prompting us to elaborate on future research directions. Building on the limitations of the present study, future work can focus on four key areas: dataset acquisition, feature set optimization, deep learning architecture development, and model interpretability. We have addressed these contents in the **Discussion** section of the manuscript.

Acquisition of dataset:

- More strategies could be explored to diversify training data beyond the current scope.
- The gas conduction and radiation can be also considered to make training data more accurate and extensive.
- For instance, large-scale experimental measurements would enable the construction of more robust and generalizable datasets.

Selection of description sets:

- Expanding the feature space to include additional material properties and environmental variables would enhance the model's ability to account for complex interactions, thereby improving its applicability across diverse scenarios.

Construction of deep learning architectures:

- Except for other convolutional neural networks (CNNs), hybrid architectures integrating models like Transformers, diffusion models, or even large language models (LLMs) could be explored for semantic feature extraction.
- Rethinking feature fusion mechanisms to explicitly integrate material/environmental factors with surface characteristics before feature identification would better simulate real-world conditions where external parameters dynamically influence surface behavior.

Visualization of models:

- More interception and visualization methods could be employed.
- Combining visualization techniques with physical theories could demystify the model's decision-making process, thereby increasing trust in its reliability.

These directions aim to address the current study's limitations, particularly its dependency on specific datasets and parameters, while expanding the framework's versatility and transparency. We believe these avenues will significantly advance the field and enable broader practical applications.

Reviewer #2:

This research predicts thermal contact resistance (TCR) through the development of a deep learning model utilizing convolutional neural networks (CNN). By generating a dataset based on surface fractal theory and multi-point contact mechanics, the authors examine the intricate relationship between surface characteristics and TCR. This study bridges theoretical modeling, supported by neural network analysis, with practical experimental applications.

Here are a few questions of this work.

- The process for generating the dataset based on surface fractal theory and multi-point contact mechanics could be elaborated. Could the authors provide a more detailed explanation of the assumptions made and the specific conditions applied during dataset generation?

We thank the reviewer for their valuable feedback and for requesting a more detailed explanation regarding the dataset generation process based on surface fractal theory and multi-point contact mechanics. We have taken several steps to address these concerns, as outlined below:

Theoretical Model:

The theoretical model has been included in the **Methods** section of the manuscript. Since this model is not original to our work, we have focused on providing the essential computational formulas necessary for reproducing the results rather than detailing the full derivation process. We believe that these formulas will allow readers to perform the required calculations independently.

Assumptions Made:

- **Environmental Conditions:** We assume that the environment is under vacuum conditions, thereby neglecting heat transfer through air gaps. This assumption simplifies the thermal analysis by focusing solely on solid-to-solid contact.
- **Temperature Distribution:** Given the relatively low overall temperature distribution in our study, we have ignored radiation heat transfer. This assumption is justified as radiative effects would be minimal under the specified conditions.

Surface Model Generation

Detailed information on the generation of fractal surfaces has been provided in the **Supplementary Method 3**. This includes specific conditions and assumptions applied during the dataset generation process.

Assumptions:

- For convenience, we assume that both contacting surfaces have the same fractal dimension. This assumption ensures two rough surfaces can be easily convert to one rough surface and one completely smooth surface. This is discussed detailly in **Supplementary Method 3**.

These details are now clearly articulated in the manuscript and supplementary materials, ensuring transparency and reproducibility. We hope this elaboration provides the clarity needed and addresses the reviewer's concerns effectively.

We appreciate the reviewer's attention to these important aspects of our methodology and welcome any further suggestions or queries they may have.

- While the model has been validated with experimental data, it would be helpful to include a comparison with existing predictive models. How does the proposed model's performance compare to conventional approaches reported in the literature?

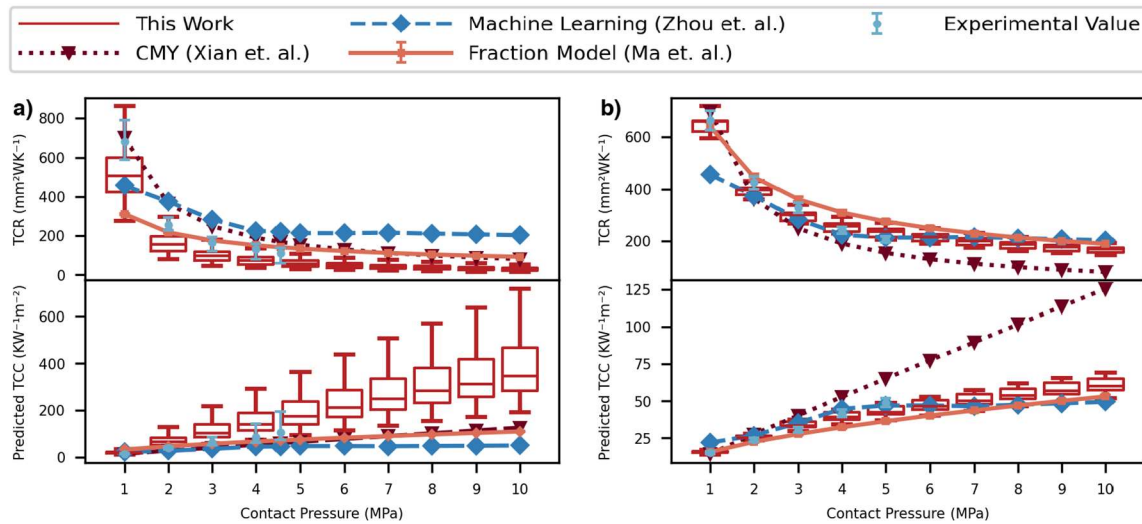
We sincerely appreciate the reviewer's insightful comment regarding the importance of comparing our proposed model with existing predictive models in the literature. In response, we have conducted a comprehensive comparison between our **deep learning** (DL) model and several conventional approaches, including the **CMY model** [1], **fractal-based model**, and a **machine learning** (ML) model [2].

As shown in **Figure 2** included in the revised manuscript and **Response Figure 2**, the predicted TCR values from our DL model are compared with experimental data as well as predictions from the aforementioned models. The detailed analysis is as follows:

1. Due to their reliance on simplified assumptions regarding surface topography, the **CMY** model and **ML** model fails to differentiate between the two sample groups. Therefore, they product the same prediction results for two kinds of specimens. In contrast, our proposed **DL** model, which directly processes surface topography images, and the **fractal model**, which accounts for surface complexity through fractal parameters, are both able to distinguish between the two sets of surfaces despite their similar roughness.
2. The first specimens:
 - **CMY**: Predictions are consistently slightly larger than experimental results.
 - **ML**: Exhibits significant deviations at all pressure levels.
 - **Fractal Model**: More consistent with predicted values within 2–4.55 MPa, but deviates from experimental values at 1 MPa.
 - **DL**: Predicts results lower than other methods on average.
3. The second specimens:
 - **CMY**: Completely fails when contact pressure exceeds 2 MPa.
 - **ML**: More consistent with predicted values within the 2–5 MPa range.
 - **Fractal Model**: More consistent with predicted values at 1–3 MPa, while being slightly larger under other conditions.
 - **DL**: Overall close to the predictions of other models, except for the **CMY** model.

We have added this comparative analysis and the corresponding figure to the **Results** section of the manuscript to better validation of our model.

- [1]. Xian, Y., Zhang, P., Zhai, S., Yang, P. & Zheng, Z. Re-estimation of thermal contact resistance considering near-field thermal radiation effect. Appl Therm Eng 157, 113601 (2019). <https://doi.org/10.1016/j.applthermaleng.2019.04.011>
- [2]. Man Zhou, Pei Yao Guo, Zhuo Yan He, Ping Zhang & Tao, L. The Thermal Contact Resistance Dataset and the Artificial Intelligence-Driven Prediction of Thermal Contact Resistance in Multi-material Systems. Chinaxiv e-prints, ChinaXiv:202504.200193 (2025). <https://chinaxiv.org/abs/202504.00185>



Response Figure 2. The prediction results compare with experiment, CMY model, ML model, and fractal model prediction for **a)** turned surfaces and **b)** ground surfaces.

- Among ResNet, DenseNet, and VGG, why did the authors choose DenseNet121?

Our selection of DenseNet121 as the model demonstrates no particular preference. In fact, to streamline our workflow, we initially planned to experiment with all built-in vision models available in PyTorch. DenseNet121 was prioritized for preliminary testing due to its alphabetical order in the model nomenclature. Fortunately, it exhibited promising performance during initial validation, which led to its adoption as the feature extractor in this study. For comprehensive architectural details of the implemented model, please refer to the complete source code we have published on GitHub.

- The authors highlight the computational efficiency of their model post-training compared to finite element simulations. It would be beneficial to quantify these efficiencies. How much faster is the prediction process, and what specific resource savings have been achieved?

We thank the reviewer for their valuable question regarding the computational efficiency of our deep learning (DL) model compared to finite element method (FEM) simulations. To address this quantitatively, we provide the following details:

- **Prediction Speed:** For the surface topography dataset of size 1024×1204, the FEM requires approximately 7 hours to complete a single prediction of TCR[1]. In contrast, after training, our DL model achieves one-second inference time per prediction,

representing a 25,200-fold acceleration (7 hours = 25,200 seconds). This stark difference underscores the model’s real-time applicability for rapid TCR estimation.

- **Training Cost:** The DL model was trained in under 100 epochs, with computational resources detailed in **Response Table 2**. While training is an upfront cost, it is a one-time expense. As the number of predictions increases, this cost is distributed across all future inferences.
- **Operational Efficiency:** Each additional prediction using the trained DL model incurs negligible computational overhead (1 second), whereas FEM retains a fixed 7-hour cost per prediction. For example, after 100 predictions, the average cost per DL prediction drops to 0.01% of the FEM cost (1 second vs. 7 hours).

The DL framework not only matches FEM’s capability to account for surface topography but also achieves orders-of-magnitude faster inference. Even when including training costs, the amortized cost per prediction becomes progressively favorable for the DL model with increasing usage. This scalability is critical for industrial applications requiring repeated TCR estimations under varying conditions.

These results are fully documented in **Supplementary Discussion 1**, which details the computational configurations for both methods. We appreciate the reviewer’s focus on quantifying efficiency and believe these metrics robustly demonstrate the model’s practical advantages.- Including a description of the mechanical conditions under which contact resistance was modeled would enhance clarity.

Response Table 2. Detailed device Specifications and computational speed comparison for FEM and DL approaches.

	FEM	DL(Inference)	DL(Train)
CPU ¹	Intel® Xeon® W-2295@3.00GHz	Intel® Xeon® E5-2680 v2@2.80GHz	AMD® EPYC™ 7J13@2.5GHz
RAM	256 GB	64 GB	512 GB
GPU	3060	4060Ti	4090
VRAM	8 GB	16 GB	24 GB
Platform	Abaqus ²	PyTorch	PyTorch
Operation System	Windows 10	Ubuntu 20.04	Ubuntu 22.04
Time Taken	~7 hour/case	~ 1 s/case	~2.5 hour/epoch ³

¹Each device is equipped with two CPUs.

²GPU acceleration was employed.

³This is the data of this paper. In fact, the vast majority of time was spent on dataset loading. This issue was systematically addressed and optimized in our subsequent work, ultimately achieving a training speed of ~1 hour/epoch.

[1]. Dong, Y., Zhang, P., Chen, M. & Lian, W. Numerical study of thermal contact resistance considering spots and gap conduction effects. Tribology International 193, 109304 (2024). <https://doi.org/10.1016/j.triboint.2024.109304>

- In Figures 5 and 6, more detailed experimental results should be presented to validate the neural network's approach.

We sincerely thank the reviewer for their constructive feedback regarding Figures 5 and 6.

In response to the comment, we would like to clarify the following:

- Regarding Figure 5 (now revised as **Figure 2**): We have indeed included more detailed experimental results in this figure to better validate the performance of the neural

network. These results demonstrate the model's accuracy in predicting thermal contact resistance (TCR) across a range of surface conditions, thereby supporting the reliability and generalization capability of our deep learning approach.

- Regarding Figure 6 (now revised as **Figure 3**): We acknowledge the reviewer's concern about the lack of direct experimental validation for this figure. Due to the current testing equipment limitations and the high integration level of the test specimens, it is not feasible to rotate the actual samples for further experimental verification.

We adopted the rotation-based data augmentation and evaluation strategy from the broader field of image recognition, where rotational invariance is a commonly assessed property. This approach allows us to evaluate the model's ability to maintain consistent prediction performance under different orientations of input surface topography images. Our intention was to illustrate that the model exhibits robustness to such variations, which is an important characteristic for image-based predictive models.

If the reviewer believes that only experimentally validated results should be included in the manuscript, we are happy to **remove** the related discussion on rotational consistency in the revision to ensure all presented findings are fully supported by experimental data, and the rotational analysis would be discussed in the **supplementary**.

This response is polite, clear, and offers a reasonable justification while remaining open to the reviewer's preferences regarding experimental validation. - The authors used stainless steel 316 exclusively for validation. How would the analysis hold if the material were changed? Is the approach still valid for other materials?

We thank the reviewer for their insightful question regarding the generalizability of our analysis to other materials. In the current study, we focused on stainless steel 316 for validation, and indeed, the framework as presented requires the generation of new datasets and retraining of the model when applied to different materials. This limitation is explicitly discussed in the **Discussion** section of the manuscript.

However, we emphasize that the associated costs of retraining are practical and feasible within engineering workflows, as elaborated in the **Supplementary Discussion 1**. The modular design of our approach ensures that the framework can be systematically extended to incorporate additional materials or parameters (e.g., material-specific properties) in future studies.

We believe these revisions significantly strengthen the rigor and clarity of the work. Please do not hesitate to contact us if additional modifications are required.

Once again, thank you for your valuable guidance.

Sincerely,

Zhang, Ping

On behalf of all authors